RELATIVE INDEPENDENCE OF ERDŐS PROBLEM #501

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ABSTRACT. We prove a strengthening of the positive assertion in Erdős problem #501 assuming that Lebesgue measure extends to a measure on all subsets of $\mathbb{R}$. Together with Hechler's counterexample under the continuum hypothesis, this gives a relative independence result for Erdős problem #501, relative to the consistency of this extension hypothesis, and hence relative to the consistency of a measurable cardinal.

1. INTRODUCTION

Let m denote Lebesgue measure on $\mathbb{R}$, and let m^* denote Lebesgue outer measure. Erdős problem #501 [1] asks whether every family $(A_x)_{x\in\mathbb{R}}$ of bounded subsets of $\mathbb{R}$ with $m^*(A_x) < 1$ admits an infinite independent set, that is, an infinite set $X \subseteq \mathbb{R}$ such that $x \notin A_y$ whenever $x, y \in X$ are distinct. We write P for the affirmative answer to this question.

Erdős and Hajnal [2] proved the existence of arbitrarily large finite independent sets. Hechler [7] proved $\neg P$ under the assumption CH. Newelski, Pawlikowski, and Seredyński [8] proved that if all sets A_y are closed and have measure < 1, then there is an infinite independent set. However, it remained open whether $\neg P$ can be proved without assuming CH.

Let FMEA denote the Full Measure Extension Axiom, the assertion that Lebesgue measure has an extension to a measure on all subsets of $\mathbb{R}$. Our main result is the following.

Theorem 1.1. *Assume* $\mathsf{ZFC} + \mathsf{FMEA}$. *Every family* $(A_y)_{y\in\mathbb{R}}$ *of subsets of* $\mathbb{R}$ *with Lebesgue outer measure* < 1 *admits an infinite independent set.*

Note that Theorem 1.1 does not assume boundedness. Hence Theorem 1.1 implies P under FMEA. FMEA is equiconsistent with the existence of a measurable cardinal; see [6, 1D(e), 2E]. Combining Theorem 1.1 with Hechler's theorem gives the following consequence.

Corollary 1.2. *If* $\mathrm{Con}(\mathsf{ZFC} + \mathsf{FMEA})$ *holds, then* P *is independent of* ZFC. *In particular, if* ZFC *with a measurable cardinal is consistent, then* P *is independent of* ZFC.

The key ingredient of the proof is Theorem 3.1, which gives a Fubini-type inequality for arbitrary subsets of $\mathbb{R}^2$. This makes it possible to handle arbitrary sets A_y, without assuming, for instance, that they are closed.

Notation. For a set X, we write $\mathcal{P}(X)$ to denote its power set. $\mathbb{R}^{<\omega}$ denotes the set of all finite sequences of real numbers; a sequence $s \in \mathbb{R}^{<\omega}$ of length n is a function from $\{0, \dots, n-1\}$ to $\mathbb{R}$.

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2. Proof of the Main Theorem

This section proves Theorem 1.1 from Lemma 2.1. We first state Lemma 2.1, postponing its proof to a later section, and then show how Theorem 1.1 follows from it.

If $\nu : \mathcal{P}(\mathbb{R}) \to [0,\infty]$ is a measure extending Lebesgue measure, then every set $S \subseteq \mathbb{R}$ satisfies

$$\nu(S) \leq m^*(S). \tag{1}$$

Indeed, this follows by covering S by countably many open intervals and taking the infimum over all such covers. In particular, if $m^*(S) < 1$, then $\nu(S) < 1$.

For a family $(A_y)_{y\in\mathbb{R}}$ of subsets of $\mathbb{R}$, write $B_x = \{y \in \mathbb{R} : x \in A_y\}$ for $x \in \mathbb{R}$.

Lemma 2.1. *Let $\nu : \mathcal{P}(\mathbb{R}) \to [0,\infty]$ be a measure extending Lebesgue measure. Let $(A_y)_{y\in\mathbb{R}}$ be a family of subsets of $\mathbb{R}$ such that $m^*(A_y) < 1$ for every $y \in \mathbb{R}$. If $C \subseteq \mathbb{R}$ and $\nu(C) = \infty$, then there is an $x \in C$ such that $\nu(C \setminus B_x) = \infty$.*

Proof of Theorem 1.1. Assume FMEA, and fix a measure $\nu : \mathcal{P}(\mathbb{R}) \to [0,\infty]$ extending Lebesgue measure. Let $(A_y)_{y\in\mathbb{R}}$ satisfy $m^*(A_y) < 1$ for every $y \in \mathbb{R}$. Fix a well-ordering $\preceq$ of $\mathbb{R}$ and a point $r_0 \in \mathbb{R}$.

For a finite sequence $s \in \mathbb{R}^{<\omega}$, define

$$C_s = \mathbb{R} \setminus \bigcup_{i<|s|} \left(A_{s(i)} \cup B_{s(i)} \cup \{s(i)\}\right), \tag{2}$$

where $|s|$ denotes the length of s. Let

$$Q_s = \{a \in C_s : \nu(C_s \setminus B_a) = \infty\}. \tag{3}$$

Define $G : \mathbb{R}^{<\omega} \to \mathbb{R}$ by

$$G(s) = \begin{cases} \text{the } \preceq\text{-least element of } Q_s, & Q_s \neq \varnothing, \\ r_0, & Q_s = \varnothing. \end{cases}$$

We write $f \restriction n$ for the restriction of f to $\{0,\ldots,n-1\}$. By recursion on ω, there exists $f : \omega \to \mathbb{R}$ such that

$$f(n) = G(f \restriction n) \qquad (n < \omega).$$

We prove by induction that

$$\nu(C_{f\restriction n}) = \infty \qquad (n < \omega).$$

For $n = 0$, $f \restriction 0$ is the empty sequence, so $C_{f\restriction 0} = \mathbb{R}$ and $\nu(C_{f\restriction 0}) = \infty$. Suppose $\nu(C_{f\restriction n}) = \infty$. Lemma 2.1 applied to $C_{f\restriction n}$ and (3) show that $Q_{f\restriction n}$ is nonempty. Hence $f(n) = G(f \restriction n)$ belongs to $C_{f\restriction n}$ and satisfies

$$\nu(C_{f\restriction n} \setminus B_{f(n)}) = \infty.$$

Moreover, by (2),

$$C_{f\restriction(n+1)} = C_{f\restriction n} \setminus \left(A_{f(n)} \cup B_{f(n)} \cup \{f(n)\}\right) = (C_{f\restriction n} \setminus B_{f(n)}) \setminus \left(A_{f(n)} \cup \{f(n)\}\right).$$

By (1),

$$\nu(A_{f(n)}) \leq m^*(A_{f(n)}) < 1,$$

and $\nu(\{f(n)\}) = 0$ because ν extends Lebesgue measure. Thus $\nu(A_{f(n)} \cup \{f(n)\}) < \infty$. Since $\nu(C_{f\restriction n} \setminus B_{f(n)}) = \infty$ and $\nu(A_{f(n)} \cup \{f(n)\}) < \infty$, we have $\nu(C_{f\restriction(n+1)}) = \infty$. This completes the induction.

Set $X = \{f(n) : n < \omega\}$. Let $i < j$. By (2),

$$f(j) \in C_{f\restriction j} \subseteq C_{f\restriction(i+1)} = C_{f\restriction i} \setminus \left(A_{f(i)} \cup B_{f(i)} \cup \{f(i)\}\right),$$

we have $f(j) \neq f(i)$, $f(j) \notin A_{f(i)}$, and $f(j) \notin B_{f(i)}$. The last relation is equivalent to $f(i) \notin A_{f(j)}$. Thus X is infinite, and $x \notin A_y$ for all distinct $x, y \in X$. ☐

3. Proof of Lemma 2.1

We use the following theorem of Kunen [3, 543C]. For a topological space X, $w(X)$ is the least cardinality of a base for the topology of X. For a measure ν, define $\mathrm{add}(\nu)$ as in [4, 511G(a)].

Theorem 3.1. *Suppose that $(Y, \mathcal{P}(Y), \nu)$ is a σ-finite measure space and that $(X, \mathcal{T}, \Sigma, \mu)$ is a σ-finite quasi-Radon measure space*[1] *with $w(X) < \mathrm{add}(\nu)$. Then every function $F : X \times Y \to [0, \infty]$ satisfies*

$$\overline{\int_X} \int_Y F(x,y)\, d\nu(y)\, d\mu(x) \le \int_Y \overline{\int_X} F(x,y)\, d\mu(x)\, d\nu(y). \tag{4}$$

For the measure ν in Lemma 2.1, $(\mathbb{R}, \mathcal{P}(\mathbb{R}), \nu)$ is σ-finite and $(\mathbb{R}, \mathcal{L}, m)$ is a σ-finite quasi-Radon measure space, where $\mathcal{L}$ denotes the σ-algebra of Lebesgue measurable subsets of $\mathbb{R}$. Moreover, $\mathbb{R}$ has a countable base, while countable additivity of ν gives

$$w(\mathbb{R}) = \omega < \mathrm{add}(\nu).$$

Applying Theorem 3.1 to the indicator of any $H \subset \mathbb{R}^2$, we get

$$\overline{\int_{\mathbb{R}}} \nu(H_x)\, dm(x) \le \int_{\mathbb{R}} m^*(H^y)\, d\nu(y), \tag{5}$$

where

$$H_x = \{y \in \mathbb{R} : (x,y) \in H\}, \qquad H^y = \{x \in \mathbb{R} : (x,y) \in H\}.$$

Here $\overline{\int_{\mathbb{R}}} g\, dm$ denotes the Lebesgue upper integral; that is, for $g : \mathbb{R} \to [0, \infty]$,

$$\overline{\int_{\mathbb{R}}} g\, dm = \inf\left\{\int_{\mathbb{R}} h\, dm : g \le h,\ h \text{ is Lebesgue measurable}\right\}.$$

We now prove Lemma 2.1. Let $C \subseteq \mathbb{R}$ satisfy $\nu(C) = \infty$. Suppose, toward a contradiction, that

$$\nu(C \setminus B_x) < \infty, \qquad \forall x \in C. \tag{6}$$

For $N \ge 1$, put $C_N = C \cap [-N, N]$. Since $C = \bigcup_{N \ge 1} C_N$ and the sequence $(C_N)_{N \ge 1}$ is increasing, continuity from below gives

$$\nu(C_N) \longrightarrow \infty. \tag{7}$$

Choose N such that $D := C \cap [-N, N]$ satisfies $\nu(D) > 1$. For $k \ge 1$, define

$$D_k = \{x \in D : \nu(C \setminus B_x) \le k\}. \tag{8}$$

By (6), $D = \bigcup_{k \ge 1} D_k$, and the sequence $(D_k)_{k \ge 1}$ is increasing. Again by continuity from below, there is $k \ge 1$ such that $\nu(D_k) > 1$. By (1),

$$m^*(D_k) > 1. \tag{9}$$

[1] See [5, 411H(a)] for the definition.

Also $D_k \subseteq [-N, N]$, so

$$m^*(D_k) < \infty. \tag{10}$$

For $M > N$,

$$\nu(C_M) \le \nu([-M, M]) = 2M < \infty.$$

By (7), let $M > N$ be large enough that $\nu(C_M) > k$.

If $x \in D_k$, then $\nu(C \setminus B_x) \le k$ by (8). Since C_M has finite ν-measure and $C_M \setminus B_x \subseteq C \setminus B_x$, we have

$$\nu(B_x \cap C_M) = \nu(C_M) - \nu(C_M \setminus B_x) \ge \nu(C_M) - k, \qquad \forall x \in D_k. \tag{11}$$

Let

$$H = \{(x, y) \in D_k \times C_M : x \in A_y\} \subseteq \mathbb{R}^2.$$

For $x \in \mathbb{R}$,

$$H_x = \begin{cases} B_x \cap C_M, & x \in D_k, \\ \varnothing, & x \notin D_k. \end{cases}$$

Thus (11) implies

$$\nu(H_x) \ge (\nu(C_M) - k) 1_{D_k}(x) \qquad (x \in \mathbb{R}).$$

By monotonicity of upper integral and the identity

$$\overline{\int_{\mathbb{R}}} a 1_S \, dm = a\, m^*(S) \qquad (a \ge 0,\ S \subseteq \mathbb{R}),$$

we get

$$\overline{\int_{\mathbb{R}}} \nu(H_x) \, dm(x) \ge (\nu(C_M) - k) m^*(D_k). \tag{12}$$

For $y \in \mathbb{R}$,

$$H^y = \begin{cases} A_y \cap D_k, & y \in C_M, \\ \varnothing, & y \notin C_M. \end{cases}$$

Since $m^*(A_y) < 1$ for every y,

$$m^*(H^y) \le 1_{C_M}(y) \qquad (y \in \mathbb{R}).$$

The function $y \mapsto m^*(H^y)$ is ν-measurable because ν is defined on all subsets of $\mathbb{R}$. Hence

$$\int_{\mathbb{R}} m^*(H^y) \, d\nu(y) \le \int_{\mathbb{R}} 1_{C_M}(y) \, d\nu(y) = \nu(C_M). \tag{13}$$

Applying (5) to this H and using (12) and (13), we obtain

$$(\nu(C_M) - k) m^*(D_k) \le \nu(C_M).$$

Dividing by $\nu(C_M) > 0$ gives

$$\left(1 - \frac{k}{\nu(C_M)}\right) m^*(D_k) \le 1.$$

Letting $M \to \infty$ through integers greater than N and using (7), together with (10), yields $m^*(D_k) \le 1$. This contradicts (9). Therefore (6) is false, and some $x \in C$ satisfies $\nu(C \setminus B_x) = \infty$. This proves Lemma 2.1.

Appendix A. Counterexample Under CH

For completeness, we give a simple proof that P fails under CH. Assume CH, and fix an enumeration $\mathbb{R} = \{r_\alpha : \alpha < \omega_1\}$. Let $\preceq$ be the induced well-ordering of $\mathbb{R}$, so that $r_\alpha \preceq r_\beta$ if and only if $\alpha \leq \beta$, and let $\prec$ be the corresponding strict order. For every $\beta < \omega_1$, the set $\{r_\alpha : \alpha < \beta\}$ is countable. For $y = r_\beta$, define

$$A_y = \{r_\alpha : \alpha < \beta,\ |r_\alpha| \leq |y| + 1\}.$$

Then A_y is countable, and hence $m^*(A_y) = 0$. Also $A_y \subseteq [-(|y|+1), |y|+1]$, so A_y is bounded.

We claim that the family $(A_y)_{y\in\mathbb{R}}$ has no infinite independent set. Suppose otherwise, and let $X \subseteq \mathbb{R}$ be infinite and independent. Choose a strictly increasing sequence

$$x_0 \prec x_1 \prec x_2 \prec \cdots$$

from X. If $i < j$, then independence gives $x_i \notin A_{x_j}$. Since $x_i \prec x_j$, the definition of A_{x_j} implies

$$|x_i| > |x_j| + 1.$$

In particular, $|x_n| > |x_{n+1}| + 1$ for every $n < \omega$. Iterating, we get $|x_n| < |x_0| - n$, which is impossible for $n > |x_0|$. Therefore no infinite independent set exists, and P fails under CH.

Use of AI

During the preparation of this work, the author used OpenAI's GPT-5.5 Pro to explore possible proof strategies and to obtain the main methodological ideas behind the argument. After using this tool, the author independently verified the mathematical details, refined the exposition, and takes full responsibility for the final manuscript.

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